

Shared Control of a BCI-Driven Wheelchair

Stage 1 Report: EEG Decoding and Baseline Characterisation

Pragyan Mohanty — Roll No. 24250 — Control Systems Project

Mid-Semester Submission, September 2026

Abstract

A motor-imagery brain-computer interface (BCI) is an unreliable control channel. It is often wrong, it responds late, and it updates slowly. This project studies how control authority should be shared between a BCI user and an autonomous navigator so that a wheelchair remains stable and safe. Stage 1 establishes the empirical basis for that study. A conventional CSP-LDA decoder was trained and evaluated across recording sessions on BCI Competition IV Dataset 2a. Over nine subjects it reaches a mean accuracy of 0.443 against a chance level of 0.25. Calibration reduces the expected calibration error from 0.196 to 0.096, and median decision latency ranges from 2 to 4 s. These parameters are now frozen and define the channel model for Stage 2.

1 Problem Statement and Objective

Motor-imagery BCIs decode imagined limb movements from scalp EEG and give people with severe motor impairment a way to issue commands. As a control input, however, the decoded signal is a degraded measurement of user intent. It has a high error rate, a transport delay of several seconds, and an update rate limited by the decoder's decision window. Closing a feedback loop directly through such a channel compromises both stability and safety.

The objective is to design and analyse a shared-control law that allocates authority between the user and an autonomous navigator, keeping the closed loop stable under transport delay and safe under decoder error while preserving as much user authority as possible. The work is divided into two stages. Stage 1 characterises the decoder from real data. Stage 2 designs and optimises the control law around that characterisation. The order is deliberate: the disturbance model used in Stage 2 must be measured, not assumed.

2 Proposed Solution

The actuator command is a convex combination of the decoded user command and the autonomous command,

$$u(t) = \alpha(t) u_{\text{BCI}}(t) + (1 - \alpha(t)) u_{\text{auto}}(t), \quad \alpha \in [0, 1], \quad (1)$$

with the authority adapted online from the decoder's calibrated confidence c and a situational risk signal r ,

$$\alpha = c(1 - r). \quad (2)$$

For fixed α and total loop delay τ , one axis of the blended loop has the characteristic equation

$$\Delta(s) = s^2 + d s + (1 - \alpha)(K_1 + K_2 s) + \alpha(K_p + K_d s) e^{-\tau s} = 0, \quad (3)$$

where K_1, K_2 are the LQR gains and K_p, K_d the operator gains. The loop is stable for any delay when $\alpha < \alpha^* = K_1/(K_1 + K_p)$.

Framework

Concept	Role in the project
EEG decoding	Common Spatial Patterns and shrinkage LDA on the 8–30 Hz band (μ and β rhythms)
Probability calibration	Platt scaling; quality measured by expected calibration error (ECE). Supplies the confidence c
Stochastic channel model	Two-state Markov error process (reliable/degraded) with measured latency and confusion structure
Plant dynamics	Unicycle wheelchair with first-order actuator lag; damped double integrator $\dot{p} = v$, $\dot{v} = u - d v$ for analysis
Operator model	Delayed PD controller, $u_h(t) = K_p e(t - \tau_h) - K_d v(t - \tau_h)$
Optimal control	LQR, with gains from the algebraic Riccati equation
Obstacle avoidance	Dynamic Window Approach over the velocities reachable under actuator limits
Situational risk	Time-to-collision mapped to $r \in [0, 1]$ and low-pass filtered
Authority conditioning	Schmitt-trigger hysteresis and slew-rate limiting on α
Time-delay stability	Substituting $s = j\omega$ removes τ from the magnitude condition, leaving a quadratic in ω^2 and a closed-form delay margin $\tau_{\max}(\alpha)$
Verification	Padé (2,2) comparison, DDE simulation and bisection, the <code>control</code> package, and MATLAB
Statistics	Paired Monte-Carlo runs with subject-level bootstrap confidence intervals

3 Dataset

The study uses BCI Competition IV Dataset 2a (BNCI2014_001), accessed through MOABB and MNE-Python: nine subjects, 22 EEG channels at 250 Hz, and four imagined movements (left hand, right hand, feet, tongue), mapped to turn left, turn right, forward and stop. Each subject was recorded in two sessions on different days, with 288 balanced trials per session. The decoder is trained on the first session and tested only on the second. This cross-session split is stricter than within-session cross-validation and closer to real use.

4 Stage 1 Method

Preprocessing. Fourth-order Butterworth band-pass at 8–30 Hz, EOG channels removed, epochs from 0.5 to 2.5 s after the cue, no baseline correction.

Decoder. Six CSP components with log-variance features, classified by LDA with Ledoit–Wolf shrinkage. The pipeline is intentionally standard, since the project needs a representative decoder rather than an optimal one.

Calibration. Platt scaling fitted on a stratified 25% hold-out drawn from the training session. Calibration matters here because the controller uses confidence to set α , and an overconfident decoder would hand authority to wrong commands.

Decision latency. The dataset is cued, so latency was measured directly. The decoder’s 2 s window is slid over 0–4 s after the cue in 0.1 s steps, and latency is the end of the first window in which the calibrated posterior exceeds 0.5. Trials that never cross are kept and censored at 4 s.

Label handling. MNE returns event codes in alphabetical order rather than the order given in the dataset documentation. Labels were therefore resolved by name, and this is covered by a unit test.

5 Results

Table 1: Cross-session decoder performance. Chance accuracy is 0.25.

Subject	Accuracy	κ	ECE (cal.)	ECE (raw)	Median latency (s)	Censored
1	0.556	0.407	0.150	0.158	2.00	0.12
2	0.556	0.407	0.075	0.167	3.55	0.46
3	0.691	0.588	0.045	0.164	2.00	0.02
4	0.389	0.185	0.077	0.131	4.00	1.00
5	0.191	-0.079	0.144	0.268	4.00	0.92
6	0.378	0.171	0.030	0.190	4.00	0.83
7	0.396	0.194	0.188	0.371	2.00	0.02
8	0.497	0.329	0.094	0.122	3.70	0.40
9	0.337	0.116	0.063	0.192	4.00	0.99
Mean	0.443	0.258	0.096	0.196	—	—

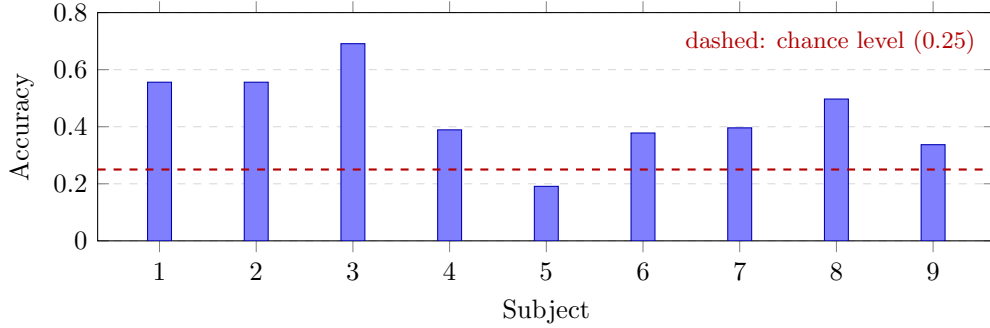


Figure 1: Per-subject accuracy on the unseen session.

Accuracy. The decoder is correct 44.3% of the time, so it is wrong on more than half of all decisions. Performance varies widely between subjects, from 0.691 (subject 3) down to 0.191 (subject 5, below chance). Subject 5 is retained on purpose, as a below-chance decoder is the case a shared-control law must handle. Because the classes are balanced, $\kappa = (\text{acc} - 0.25)/0.75$ exactly, so κ is reported only for comparison with the literature.

Comparison with published results. Mean κ (0.258) is below the value of about 0.4 commonly reported for this dataset. Data leakage was ruled out: for subjects 4 and 9 the held-out session scores higher than within-session cross-validation, which leakage would not produce. Subject 3 reaches $\kappa = 0.784$ under cross-validation, confirming that the pipeline works. The gap reflects signal drift between recording days. Higher published figures typically rely on within-session evaluation or filter-bank CSP.

Calibration. Platt scaling reduced the mean ECE from 0.196 to 0.096 and improved every subject. The calibrated confidence is informative but weak. Across all 2592 test trials it averages 0.514 on correct decisions and 0.408 on incorrect ones (ROC AUC 0.676), and within individual subjects the separation is smaller.

Latency. Subjects 1, 3 and 7 typically become confident within the first window (median 2.0s). For subjects 4, 5, 6 and 9, 83–100% of trials never reach the threshold within 4s. Some subjects show a bimodal distribution, so latency will be resampled empirically in Stage 2 rather than fitted to a parametric form.

Baseline carried forward. The decoder is now frozen. Stage 2 receives, for every test trial, the correctness of the decision, the per-subject confusion matrix, the calibrated confidence and the decision latency. Together these define the error rate, error direction, confidence term and loop delay of the simulated BCI channel.

6 Remaining Work (Stage 2)

1. Fit a per-subject Markov error model and latency distribution from the Stage 1 outputs.
2. Implement the wheelchair and double-integrator plants and the delayed-PD operator.
3. Implement the LQR and Dynamic Window Approach controllers.
4. Implement fixed, confidence-only and risk-gated adaptive arbitration, $\alpha = c(1 - r)$.
5. Derive $\tau_{\max}(\alpha)$ and α^* analytically and verify them numerically.
6. Compare all policies over 9 subjects, 3 environments and 20 seeds on success rate, collision rate and retained user authority.
7. Consolidate results, ablations and limitations in the final report.

7 Limitations of Stage 1

The EEG is recorded offline and cannot respond to the wheelchair’s motion, so it enters the loop only through measured statistics. The cued, randomised protocol may produce different error patterns from a self-paced BCI. Latency is estimated from sliding windows and is heavily censored for some subjects. Finally, the decoder is standard rather than state of the art, by design.

8 Note

This project is under active development. Stage 1 is complete and documented here. Stage 2 is in the development phase, and the source code will be uploaded to the repository after the mid-semester evaluation.

References

- [1] C. Brunner et al., *BCI Competition 2008 – Graz data set A*, Graz University of Technology, 2008.
- [2] V. Jayaram and A. Barachant, “MOABB: trustworthy algorithm benchmarking for BCIs,” *J. Neural Eng.*, 15(6), 2018.
- [3] A. Gramfort et al., “MEG and EEG data analysis with MNE-Python,” *Front. Neurosci.*, 7:267, 2013.
- [4] G. Pfurtscheller and C. Neuper, “Motor imagery and direct brain–computer communication,” *Proc. IEEE*, 89(7), 2001.
- [5] H. Ramoser, J. Müller-Gerking and G. Pfurtscheller, “Optimal spatial filtering of single trial EEG during imagined hand movement,” *IEEE Trans. Rehabil. Eng.*, 8(4), 2000.
- [6] J. C. Platt, “Probabilistic outputs for support vector machines,” in *Advances in Large Margin Classifiers*, MIT Press, 1999.
- [7] A. D. Dragan and S. S. Srinivasa, “A policy-blending formalism for shared control,” *Int. J. Robot. Res.*, 32(7), 2013.
- [8] D. Fox, W. Burgard and S. Thrun, “The dynamic window approach to collision avoidance,” *IEEE Robot. Autom. Mag.*, 4(1), 1997.